STUDENT REPORT

DETAILS

Name

THANGADIPALLY ANVITHA

Roll Number

20R01A6753

EXPERIMENT

Title

WEEK 11A

Description

11.Write R program to find the following Distribution with the sample data and visualize the linear regression graphically.

a) Normal Distribution

Procedure and Overvation:

20R01A6753

DESCRIPTION: In a random collections of data from independent sources it is generally observed that the distribution of data is normal which means on plotting a graph with the values of the variable in the horizontal access and the count of the values in the vertical axis we get a bell-shaped.

In the graph 50% of values lie to left of the mean and the other 50% lie to the right of the ground .this refer to a s normal distribution in stastics.

R has four in built funtions to generate normal distributio.they are described below.

SOURCE CODE:

```
dnorm(x,mean,sd)

pnorm(x,mean,sd)

qnorm(p,mean,sd)

r norm(n,mean,sd)

p-vector probabilities

->x is a vector of numbers

->p is a vector of probabilities

->n is number of observations

->mean is the value of the sample data .it's default value is zero

->sd is the standard deviation .it's default value is 1
```

DNORM:This fuction gives height of the probability distribution at each poin for a given mean and standard deviation

#create a sequence of numbers between -10&+10 increment 0.1

SOURCE CODE:

```
#Create a sequence of numbers between -10 and 10 incrementing by 0.1.
x<-seq(-10,10,by=0.1)
# Choose the mean as 2.5 and standard deviation as 0.5.
y<-dnorm(x,mean=2.5,sd=0.5)
# Give the chart file a name.
png(file="dnorm.png")
plot(x,y)
# Save the file.
dev.off()
```

PNORM:this function gives the probability normal distributed random numbers to be less that the value of a given num ber it is also called qumilative distribution function

SOURCE CODE:

```
# Create a sequence of numbers between -10 and 10 incrementing by 0.2.
x<-seq(-10,10,by=0.2)
# Choose the mean as 2.5 and standard deviation as 2.
y<-pnorm(x,mean=2.5,sd=2)
# Give the chart file a name.
png(file = "pnorm.png")
# Plot the graph.
plot(x,y)
```

```
# Save the file.
dev.off()
```

QNORM:this function take the probability value these number the qumilative value matches the probability value.

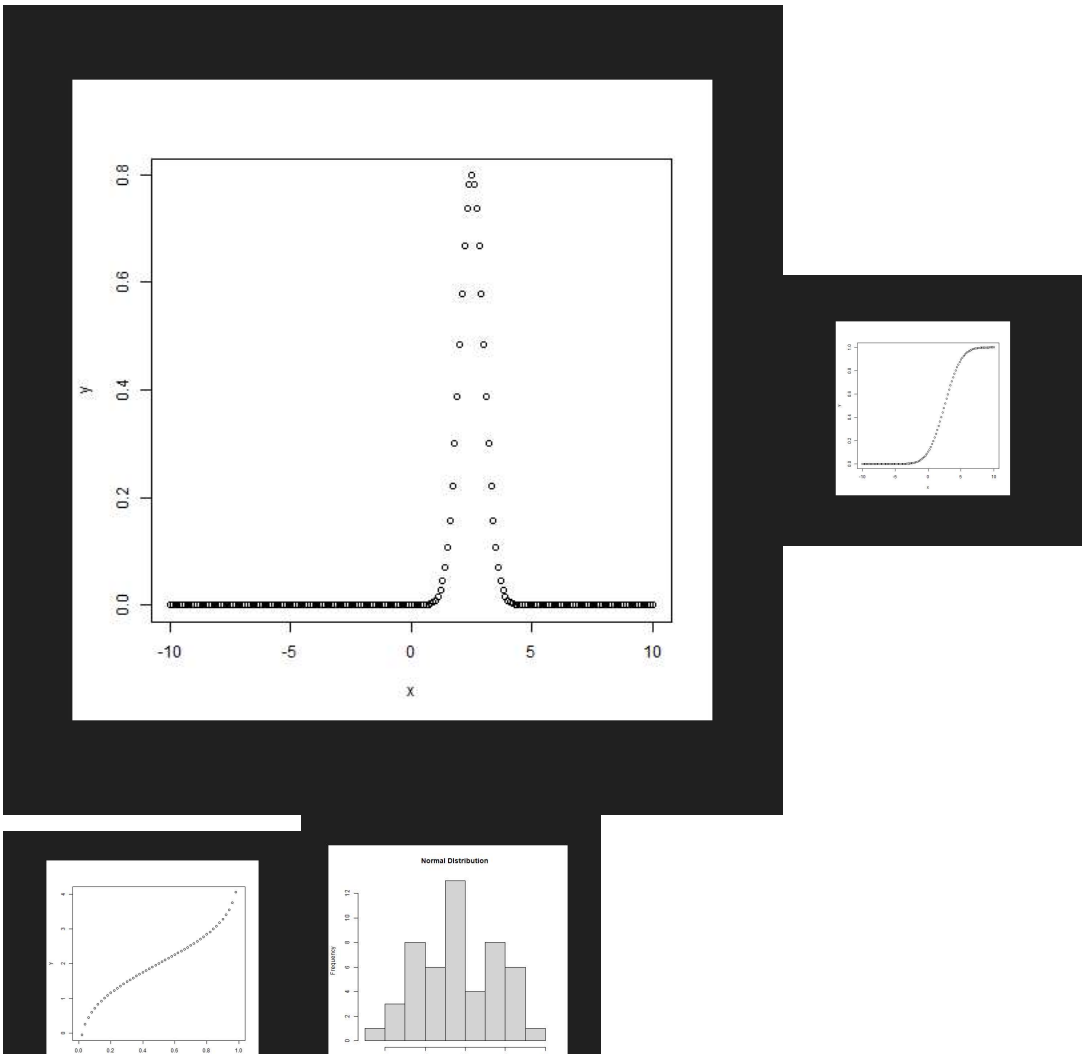
SOURCE CODE:

```
# Create a sequence of probability values incrementing by 0.02.
x<-seq(0,1,by=0.02)

# Choose the mean as 2 and standard deviation as 3.
y<-qnorm(x,mean=2,sd=1)
# Give the chart file a name.
png(file="qnorm.png")

# Plot the graph.
plot(x,y)
# Save the file.
dev.off()
```

RNORM:this function is easy to select random number whose distribution normal number the sample size is input and g enerates that many random number to draw histogram to show the distribution of the generated numbers .

SOURCE CODE:

```
# Create a sample of 50 numbers which are normally distributed.
y<-rnorm(50)
# Give the chart file a name.
png(file="rnorm.png")
# Plot the histogram for this sample.
hist(y,main="Normal DIstribution")
# Save the file.
dev.off()
```



VIVA RESPONSES

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